Prepared for referee review

NEURAL BELLMAN OPERATORS

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Neural Bellman Operators construct and learn continuation operators for feasible decisions under a specified recursive economic objective. We derive explicit regularity of the original action marginal in a constrained investment economy with continuous innovations and nested average value at risk, including binding capacities and moving adverse tails. Paid original-risk value enclosures supply outward derivative labels. A bounded action-marginal readout has an exactly checkable primal–dual training gap; feature-aware continuation contrasts and constrained action bounds state what additional work controls original policy loss. A recursive level-and-contrast theorem shows why a state-dependent critic anchor cannot be discarded when exported to an earlier Bellman operator. Complete proofs retain the own-future construction, stochastic-control, recursive-utility and boundary assumptions. The prospective independent study compares two learning losses and acquisition budgets for neural, matched-hinge, quadratic, polynomial, spline and grid critics against a strong online method, with all acquisition, refusal, qualification and complete-service costs. Its source, actor and new state-sample boundaries are fixed separately. At this source-freeze stage no new numerical outcome is asserted; the completed author manuscript will report the actually executed original-policy and cost comparisons, including adverse results and unresolved adoption regions.

KEYWORDS: Dynamic programming, Neural approximation, Recursive risk, Policy qualification, Computational economics.

1. INTRODUCTION

A dynamic decision maker buys an approximation because it changes the cost of making a good decision. In a constrained investment problem, an accurate continuation level can still give the wrong incentive to invest, while a continuation

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with a large common level error can choose the correct action. Under recursive risk, the difficulty is greater: changing a continuation can also change which shocks receive adverse weight. A numerical method must therefore connect its acquired information, its fitted critic, its feasible actor and its computational work to the original recursive objective.

Neural Bellman Operators (NBOs) address this chain by constructing and learning recursive continuation operators together with an account of their policy error. The object is the Bellman operator of the specified economy. Its criterion, information filtration, constraints and boundary conditions remain part of that object throughout approximation and deployment. The paper retains both a constructive own-future neural operator and an analysis of arbitrary acquired critics. It develops their links with stochastic control and, under separate explicit assumptions, nonlinear Brownian recursive utility. A result for the nested risk investment application is not automatically a welfare improvement in a different expected-utility or nonlinear recursive-utility model.

The central computational question is which continuation information is worth acquiring. Earlier NBO calculations found a concrete mismatch: every certificate-oriented fit in a retained comparison reduced a residual-span criterion, but the aggregate number of rejected first actions increased from 644 to 757. The quadratic control remained a strong complete-service baseline. The subsequent policy-comparison calculations preserved that economic criterion and constructed safe actual improvements, yet acquisition still used a value loss. Those results motivate a new learning calculation in the original model. Their observations and costs retain their historical identities in the development companion; the present sample does not reuse them.

We learn the marginal continuation cost after evaluating the original risk operator. This order matters. In the two-date investment economy, the original optimal continuation is the last-date Bellman image of the known terminal function. It is acquired through paid outward calculations. The root critic retains the known action cost and learns the action-dependent part of the integrated continuation. A state-only anchor completes its value level. The resulting actor minimizes that learned objective over the original capacity interval and is admitted only through a certificate for the original complete policy. The same acquired values, features where

relevant, feasible actions and certification rules are available to the conventional controls.

The mathematical contribution has three connected parts. First, we derive a continuous original-action marginal and explicit moduli from the nonlinear investment technology, binding capacity and moving AVaR tails. The adverse central interval can touch either shock boundary. Its variational inequalities control movement of the tail without differentiating a numerical sort or an unconstrained optimizer. Finite original-risk value enclosures then provide outward derivative labels, with the precision cost of finite differences paid explicitly.

Second, a bounded readout minimizes the worst action-marginal discrepancy over those outward labels. The finite training problem is convex for each declared feature family. Exact primal and dual bounds certify the actual stored coefficients without treating a floating solver's success flag as a proof. An explicit feature-range construction extends the acquired information to off-design action comparisons. Its integral along the feasible action path controls policy loss and possible policy changes; a capacity-sensitive stationarity bound sharpens the account when available. The argument applies to a nonconvex fitted polynomial or spline through the all-action comparison, and never assumes that target smoothness removes a fitted ReLU kink.

Third, we state what is lost if the critic's anchor is discarded recursively. In this very investment transition, a state-dependent continuation level that does not change a last-date action can shift the root marginal by a known nonzero amount. A full level-and-contrast recursion therefore carries the anchor error, the action gap and the future policy error separately. A noncash-invariant recursive preference receives its own continuation Lipschitz account. It cannot inherit a monetary shift identity merely because both criteria have a Bellman notation.

Derivative-assisted economic dynamic programming already uses Hermite approximation with value and slope information (**?**). Derivative supervision, task-based model learning, constrained first-order error bounds and finite-dimensional convex duality also have established precedents (**???**). The development history also contains certificate-aware readouts and an action-null quotient analysis. The additional results here concern their explicit construction and verification for the

original recursive risk operator, including tail changes, paid finite-risk information, feasible actions and recursive value levels. A three-probe lower bound identifies a limitation of that stated information interface; it is not a lower bound for every algorithm with access to the investment formulas.

The numerical comparison changes the learning loss before changing the evaluation sample. Two separately executed acquisition budgets use nested state designs. Seven stored representations each have a value-loss and an action-marginal-loss critic; the online original-model method is included at both risk levels. Quadratic, polynomial, cubic spline, multilinear grid and semi-analytic controls form the conventional menu. An exactly corresponding hinge encoding makes the neural representation claim testable: a geometry shared with that control is not a uniquely neural contribution.

The protocol fixes 64 new independent initial states, complete original-policy evaluation, five fresh matched timing processes, exact independent reconstruction, simultaneous positive and adverse gain inference and full acquisition/service cost comparisons. It also fixes how unresolved and failed calculations are retained. The empirical question is whether action-marginal learning changes decisions, rejections and complete work enough to compensate for its acquisition cost. A reduction of its own finite-node fitting objective alone does not answer that question.

The main article presents the economic operator, its original-risk qualification and the new learning construction. The supplement contains complete proofs and retains the earlier action, tail-information and policy improvement theorems. The development companion preserves the earlier applications, numerical records, negative findings and source-bound revision history. This organization keeps the active economic and mathematical argument readable while retaining the research needed to assess it.

2. THE ECONOMIC OPERATOR AND ITS POLICY ACCOUNT

Consider dates $t = 0, \dots, T$ and an invariant compact state domain $X \subset \mathbb{R}^d$. At date t, the action belongs to a nonempty compact set $A_t(x)$. The transition is $F_t(x, a, Z_{t+1})$, the current cost is $c_t(x, a)$, and the terminal cost is $g(x)$. Write $B_0 = 1$

and $B_t = \prod_{j<t} \beta_j$, with $\beta_j > 0$. An implemented policy includes its observation rule, action selection and feasible numerical realization, not just its stored parameters.

For a continuation v, define

$$Q_t(v; x, a) = c_t(x, a) + \beta_t \rho_t(v(F_t(x, a, Z))), \qquad \mathcal{T}_t v(x) = \inf_{a \in A_t(x)} Q_t(v; x, a). \tag{1}$$

Throughout the deterministic comparison, the evaluations ρ_t are monotone and cash invariant on a common domain containing all functions and constant translates used below. They are therefore nonexpansive in the supremum norm. Continuous objectives, effective compact feasible correspondences and measurable minimizers, or measurable arbitrarily near-minimizers with their error charged, are assumed. The stipulated conditional recursion gives Bellman comparison against all admissible adapted controls. In the earlier expected-cost experiments ρ_t is conditional expectation, costs are bounded and innovations have their original continuous uniform law. Section 5 evaluates the same investment technology under an explicitly nested AVaR preference and retains the original continuous innovation law. The direct-cost inference in Section BT is specific to expectation; it is not automatically an inference theorem for nonlinear recursive utility or games.

A Neural Bellman Operator is a construction and evaluation procedure for (1). It works backwards with its own acquired continuation, returns a neural critic and a feasible actor, and records an economic error allowance. Its continuation may be fitted or enclosed by an explicitly paid primitive Bellman calculation; an exact optimal continuation is never supplied as a free oracle. The constructive theorem below uses its own fitted f_{t+1} at every date. In the new two-date application, the terminal function g is an original primitive, and the paid last-date operator $T_1 g$ supplies outward target intervals shared with every conventional comparator. The deployed last-date policy retains its separately proved error. Arbitrary trained networks can be evaluated by the first result below. The constructive backend specifies how a network and its actor are obtained without assuming successful nonconvex optimization.

2.1. *Action-dependent errors and complete-policy loss*

Let $H(a)$ be a true current cost and $\widehat{H}(a)$ its continuation approximation. If $\widehat{a}$ is δ-optimal for $\widehat{H}$ on the same feasible action set, then

$$H(\widehat{a}) - \inf_a H(a) \le \delta + \operatorname{osc}_a\{\widehat{H}(a) - H(a)\}. \tag{2}$$

Indeed, compare $\widehat{a}$ with an arbitrarily near-optimal true action and cancel the approximate objectives. The infimum limit gives the result. Additive continuation levels cancel; action-dependent discrepancies do not. Under a change in future policies, the continuation must be transported to the new economic problem before this inequality licenses reuse. The full development edition retains the original quantitative transport, recursive-domain and economic-application results. No transport bound is presumed valid after an unaccounted change of constraints or preferences.

PROPOSITION 1—One-sided policy sandwich: *Suppose, uniformly in state,*

$$f_t - \mathcal{T}_t f_{t+1} \le u_t, \qquad \mathcal{T}_t^{\pi} f_{t+1} - f_t \le v_t, \qquad \ell_T \le f_T - g \le u_T.$$

Then the actual feasible policy satisfies

$$0 \le J_0^{\pi} - V_0^{*} \le \sum_{t<T} B_t(u_t + v_t) + B_T(u_T - \ell_T). \tag{3}$$

The bound is invariant to date-specific constant shifts of the fitted continuations, provided the associated residual accounts are shifted consistently.

The proof is in Supplement, Section 5.1.

A sharper valid residual account for the same continuation and the same deployed actor sharpens knowledge of the policy. It does not retrain that policy or change its realized cost. Conversely, subtracting two upper bounds in (3) does not identify the difference in actual costs. These distinctions govern both the historical comparisons and the new experiments.

3. CONSTRUCTING AND COMPILING A FEASIBLE NEURAL POLICY

Assume the primitive objective has, for feasible state–action pairs, state modulus $A_t^Q = C_{x,t} + \beta_t F_{x,t} L_{t+1}$ and action modulus $D_t = C_{a,t} + \beta_t F_{a,t} L_{t+1}$. Let

$$d_H(A_t(x), A_t(z)) \le K_t^A \|x - z\|_1, \qquad L_t = A_t^Q + D_t K_t^A, \qquad L_T = L_g.$$

The Hausdorff condition and the primitive moduli imply that $\mathcal{T}_t f_{t+1}$ is L_t-Lipschitz whenever f_{t+1} is L_{t+1}-Lipschitz. All moduli and numerical integration procedures are supplied by the economic primitives, not fitted from successful outcomes.

At state nodes $x_{t,i}$ with covering radius h_t, form effective feasible k_t-nets. Evaluate their own-future objectives with certified error at most e_t. Minimize the computed labels and retain the corresponding *original* action $a_{t,i}$. A Borel repair $\Gamma_{t,i}(x)$ is feasible at x and satisfies

$$\|\Gamma_{t,i}(x) - a_{t,i}\|_1 \le K_t^A \|x - x_{t,i}\|_1 + \zeta_t. \tag{4}$$

The allowance ζ_t concerns displacement; feasibility itself may not fail. Define

$$f_t(x) = \min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}, \quad i_t(x) = \min \arg\min_i \{y_{t,i} + L_t \|x - x_{t,i}\|_1\}. \tag{5}$$

Return $\pi_t(x) = \Gamma_{t,i_t(x)}(x)$. Terminal labels approximate g with error e_T and use the same cone construction with modulus L_g.

THEOREM 2—Own-future construction with feasible witnesses: *The functions in* (5) *have exact affine–ReLU realizations and Lipschitz moduli L_t independent of label error. The returned policy is feasible and measurable. It satisfies*

$$-e_t \le f_t - \mathcal{T}_t f_{t+1} \le D_t k_t + e_t + 2L_t h_t, \tag{6}$$
$$\mathcal{T}_t^\pi f_{t+1} - f_t \le e_t + D_t \zeta_t,$$

and hence

$$J_0^\pi - V_0^* \le \sum_{t<T} B_t (D_t k_t + 2e_t + 2L_t h_t + D_t \zeta_t) + B_T (2e_T + 2L_g h_T). \tag{7}$$

With effective covers, repairs and certified queries at the prescribed accuracy, allocating each nonnegative term a share of a positive economic tolerance gives a terminating finite

construction. A fixed numerical precision or a capped finite experimental ladder need not contain that allocation.

The proof is in Supplement, Section 5.2.

The Lipschitz-extension principle is classical (**?**). A native min-plus envelope and its exact ReLU circuit, using the same labels, witnesses and ties, are *one method with two encodings*. They return the same function and policy. Neither Theorem 2 nor a faster representation proves a uniquely neural capability or a training-success probability for Adam or L-BFGS. The mathematical increment is the complete feasible, own-future construction and implementation account; the empirical question is its work and policy value relative to conventional generators.

4. REFERENCE-CENTERED RISK KERNELS AND COMPLETE-POLICY QUALIFICATION

A datewise worst-case error account can charge incompatible states at successive dates. Under nonlinear recursive preferences, replacing that account by an expectation under the physical transition law is not valid. We instead retain the conditional risk evaluation and center its dual representation at the actual stored NBO continuation. The resulting state-dependent certificate has an original-optimum interpretation and a finite occupation representation. It extends, rather than replaces, the signed qualification of Section AP.

4.1. *The centered operator*

Retain the original augmented economic state, feasible actions, nonnegative discounts, and bounded complete-policy recursions. At each (t, x, a) suppose the original conditional evaluation admits the representation

$$\rho_{t,x,a}(v) = \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int v(y) q(dy) - \psi_{t,x,a}(q) \right\}. \tag{8}$$

Every q is a probability measure on the original successor state space. The representation is assumed on a common bounded-function domain containing all continuations below, with the measurability and selection conditions needed for the original Bellman recursions. This is a hypothesis about the original preference

primitive, not permission to replace a static risk objective by a nested one. Conditional risk duality and risk-averse dynamic programming are established principles (**??**). For a nonconvex or non-cash-invariant aggregator lacking (8), the earlier order and modulus results remain applicable.

Fix one qualified parameter θ and suppress its subscript. Its entire sequence of stored continuations is $f = (f_t)_{t=0}^{T}$, and its supplied complete feasible policy is π. Define, using the same f_{t+1} in every term,

$$\begin{aligned} r_t(x,a) &= c_t(x,a) + \beta_t \rho_{t,x,a}(f_{t+1}) - f_t(x), \\ k_t^f(x,a,q) &= \psi_{t,x,a}(q) + \rho_{t,x,a}(f_{t+1}) - \int f_{t+1}\, dq, \\ \mathcal{E}_{t,x,a}^f(z) &= \sup_{q \in \mathcal{Q}_{t,x,a}} \left\{ \int z\, dq - k_t^f(x,a,q) \right\}. \end{aligned} \tag{9}$$

The penalty is nonnegative and has infimum zero, whether or not the supremum in (8) is attained. Direct substitution gives

$$\mathcal{E}_{t,x,a}^f(z) = \rho_{t,x,a}(f_{t+1} + z) - \rho_{t,x,a}(f_{t+1}). \tag{10}$$

In particular, $\mathcal{E}^f(0) = 0$, and $\mathcal{E}^f$ is monotone, cash invariant, and nonexpansive in the uniform norm. Discarding k^f is an upper relaxation, not this identity. Freezing one risk-maximizing kernel at f is in general not even an upper relaxation.

THEOREM 3—State-dependent original-optimum certificate: *Suppose verified functions satisfy $r_t^-(x,a) \le r_t(x,a)$ for every original feasible action and $r_t(x,\pi_t(x)) \le r_t^{\pi,+}(x)$. Let bounded measurable L_t, U_t satisfy*

$$\begin{aligned} L_T &\le g - f_T \le U_T, \\ L_t(x) &\le \inf_{a \in A_t(x)} \{ r_t^-(x,a) + \beta_t \mathcal{E}_{t,x,a}^f(L_{t+1}) \}, \\ U_t(x) &\ge r_t^{\pi,+}(x) + \beta_t \mathcal{E}_{t,x,\pi_t(x)}^f(U_{t+1}). \end{aligned} \tag{11}$$

Then, at every initial state,

$$f_0 + L_0 \le V_0^* \le V_0^\pi \le f_0 + U_0, \qquad 0 \le V_0^\pi - V_0^* \le U_0 - L_0. \tag{12}$$

The same statements hold at every intervening date and state. They are uniform over a qualified parameter class whenever all premises and the reported bound hold for each member with its own common continuation sequence and complete policy. The certificate permits endogenous states and actions changing at every date; it does not require a finite state space or independent deployment contexts.

With exact residuals and terminal discrepancies, equality recursions in (11) *give* $L_t = V_t^* - f_t$ *and* $U_t = V_t^\pi - f_t$. *With scalar bounds from* (A.34), *their corresponding constant lower and upper sequences are feasible certificates here. Thus the exact state-dependent recursion is no looser than the signed scalar account. A strictly smaller bound can result without changing a fit, a feasible action, or an economic observation.*

The last assertion concerns the best exact recursion or a verified improvement upon the constant certificates. It is not a claim that every approximately computed L, U improves the scalar account. In implementation, the minimum of valid policy supersolutions and the maximum of valid optimum subsolutions retain validity, so an inferior new bound need not displace the existing one.

The lower inequality requires every original action. A discrete action search without a verified original-menu lower bound cannot establish it. For a fixed action, a feasible risk kernel supplies a lower bound on $\mathcal{E}^f(L)$; an upper bound requires control of the entire risk envelope. Numerical integration, feasible repairs, residual errors and penalty errors belong in these inequalities. If penalty enclosures are used, lower penalty endpoints enter an upper certificate and upper endpoints enter a lower witness. Approximate dual optimizers are accepted only through these independently checked inequalities.

5. CONTINUOUS-LAW QUALIFICATION IN THE INVESTMENT MODEL

The risk theorem becomes a numerical method only when its inequalities can be constructed from economic primitives. We now supply this construction for the original investment technology with a nested tail-risk preference. States and innovations remain continuous, every feasible action is covered, and the implemented terminal controller rather than a hypothetical exact optimizer determines policy cost. This application joins continuation construction, the original-optimum

comparison and a representation comparison on one problem. It does not replace the general NBO construction or the other economic applications.

5.1. *Primitives and the object being approximated*

Let $d \geq 2$ be even, $X = [0,1]^d$, and interpret subscripts cyclically. At dates $t = 0, 1$, the feasible investment is $0 \leq a \leq A(x) = 1/8 + \bar{x}/8$, where $\bar{x} = d^{-1}\sum_j x_j$. Write $b_j = 1/2$ for odd j and $b_j = 1/4$ for even j, and $s_j = (-1)^{j+1}$. The original transition is

$$F_j(x,a,z) = \frac{1}{16} + \frac{x_j}{2} + \frac{x_{j+1}}{8} + \frac{x_j(1-x_{j+1})}{16} + b_j a + s_j z, \qquad z \sim \mathrm{Unif}[-1/32, 1/32]. \tag{13}$$

Define

$$S(x) = \frac{2}{d}\sum_j (x_j - 5/8)^2 + \frac{1}{4d}\sum_j (x_j - x_{j+1})^2 + 2(1/2 - 2\bar{x})_+^2, \tag{14}$$

$$g(x) = S(x) + \frac{2}{d}\sum_j (x_j - 5/8)^2, \qquad h(a) = a^2 + 4a^4.$$

We use $\beta = 15/16$ and, separately, $\alpha \in \{1/2, 3/4\}$. Average value at risk is the upper-tail cost functional

$$\rho_\alpha(v) = \sup\left\{ \int vq\, dP : 0 \leq q \leq (1-\alpha)^{-1},\ \int q\, dP = 1 \right\}.$$

Here P is the original uniform innovation law. The objective is recursively defined by $V_2 = g$ and

$$V_t(x) = S(x) + \inf_{0 \leq a \leq A(x)} \{h(a) + \beta\rho_\alpha(V_{t+1}(F(x,a,\cdot)))\}.$$

It is not the static tail risk of total realized cost. A complete policy value replaces each infimum by the action prescribed by that policy, with the same recursive risk primitive.

Assumptions for the continuous construction. For this subsection, the primitives are exactly (13)–(14); α is fixed before construction; the fitted continuation f is a finite convex function on X; and every numerical lower or upper value is an outward

enclosure. Feasible tail densities and feasible actions are retained explicitly. Terminal selection is the measurable, dyadically acquired controller described below, with its fallback included in the error bound. These concrete conditions imply compact feasible graphs, bounded costs, measurable minimizers, and the conditional-risk dual used in Theorem 3. They do not assert that every recursive preference or every trained network has this convex structure.

PROPOSITION 4—Primitive curvature and state supports: *For either risk level and every even d, the one-date objective $Q_1(x,a) = S(x) + h(a) + \beta\rho_\alpha(g(F(x,a,\cdot)))$ is jointly convex on $X \times [0,1/4]$, with curvature bounded below by*

$$\frac{4 - 5\beta/4}{d}\|v\|_2^2 + 2w^2 \tag{15}$$

in a direction (v,w). Its value V_1 is convex on X and satisfies

$$|V_1(x) - V_1(y)| \le \frac{15}{d}\|x - y\|_1. \tag{16}$$

Let W be the objective evaluated under any fixed feasible risk density, at an anchor (x_0,a_0), and let (p_x,p_a) be its gradient there. For any $\lambda,\nu \ge 0$, set $e = p_a + \lambda - \nu$. Then the following is a global affine minorant of V_1:

$$\ell(y) = W + \lambda(a_0 - A(x_0)) - \nu a_0 - \frac{e^2}{4} + \left(p_x - \frac{\lambda}{8d}\mathbf{1}\right)^\top (y - x_0). \tag{17}$$

Neither optimal multipliers nor an exact optimizing action are required.

The curvature is obtained from the original nonlinear technology, not imposed on an interpolated value grid. The negative second-order term from $x_j(1 - x_{j+1})$ is bounded using $|\partial_j g| \le 10/d$; the stage curvature dominates it. The allowance $e^2/4$ pays for imperfect stationarity. The proof in the Technical Supplement also establishes the global state modulus in (16), including the state-dependent action constraint.

5.2. *An all-action certificate under the original innovation law*

Partition the innovation interval into n equal bins B_i, with midpoints z_i and probabilities $p_i = 1/n$. An innovation bin is a set over which an inequality holds, not a replacement point mass in the economic model. Choose nonnegative $q_i \leq p_i/(1-\alpha)$ summing to one. These masses define a feasible density constant on each bin. For a fixed initial state and feasible proposal a_0, let W_0 be a lower bound on the objective under this density, obtained from globally valid affine supports of the continuation at the successor midpoints. Let p enclose the derivative of the same supporting objective at a_0. Then

$$\underline{V}_0(x) = \inf_{0 \leq u \leq A(x)} \{W_0 + p(u - a_0) + (u - a_0)^2\} \tag{18}$$

is a lower bound on the original Bellman optimum when the supported continuation is V_1. The scalar quadratic is minimized over the full original interval. Interval uncertainty in p is covered by taking the smaller minimum at its two endpoints, since the minimum is concave in p.

For a convex fitted continuation, midpoint supports give the lower risk witness and the larger endpoint value bounds the continuation throughout each bin. For V_1, Proposition 4 supplies midpoint supports and the allowance $15/(32n)$ covers each half-bin. Upper tail risk is the average of the largest $n(1-\alpha)$ upper bin bounds. This average is the exact risk of a dominating step function, hence an upper bound under the continuous law.

THEOREM 5—Constructed original-optimum qualification: *Suppose the supports and bin bounds above are outwardly verified. Suppose further that the complete terminal controller satisfies $V_1 \leq V_1^{\widehat{\pi}} \leq V_1 + \varepsilon_1$ everywhere on X. For every queried initial state, the lower value* (18) *and the binwise upper bound $\overline{J}_0^{\widehat{\pi}}$ satisfy*

$$0 \leq J_0^{\widehat{\pi}}(x) - V_0(x) \leq \overline{J}_0^{\widehat{\pi}}(x) - \underline{V}_0(x). \tag{19}$$

For any stored convex continuation f, the same lower and upper constructions are admissible bounds for the centered-risk comparison of Theorem 3. Their validity does not depend on the representation of f, on a successful fit, or on the proposal being an exact minimizer.

To see the connection, write a verified continuation bracket as $\ell \le V_1 \le u$ and use errors $L = \ell - f$, $U = u - f$. The centered identity gives

$$\rho_\alpha(f) + \mathcal{E}^f(L) = \rho_\alpha(\ell), \qquad \rho_\alpha(f) + \mathcal{E}^f(U) = \rho_\alpha(u).$$

Joint evaluation avoids subtracting and then adding two separately rounded copies of $\rho_\alpha(f)$. The bound (18) uses one feasible risk density to minorize the required all-action infimum; the policy upper bound controls the full risk envelope. Thus the application constructs sufficient inequalities for the original theorem. It does not claim that a non-neural continuation would receive a different validity theorem, or that computing the original-value witness is free.

The terminal controller observes each state coordinate rounded down to 40 binary places. It accepts its floating proposal only after an outward all-action gap of at most 2^{-21} at that observed state; otherwise it uses rational bisection. The original-state transfer is less than $30\,2^{-40}$. Consequently $\varepsilon_1 = 2^{-20}$ covers the complete controller, including observation and fallback. Policy evaluation encloses this controller at every possible successor, not only at the midpoint states used by the integrator.

Well-posed original decisions. The comparison uses a specified conditional nested recursion on the original observation filtration. A sufficient primitive model has a compact state space, nonempty compact feasible sets varying continuously with state, continuous costs and invariant transitions, compact innovations with their original fixed laws, and finite monotone cash-invariant risk evaluations. These assumptions make the Bellman operator preserve continuous functions. For implemented Borel policies, the risk evaluation must also preserve Borel parameter kernels; the supplement proves this property for expectation and AVaR. Proposition S.9 proves, rather than assumes, the resulting finite-horizon value existence and uniqueness, attained Borel minimizers, comparison against all feasible adapted history policies, and the dated stability bounds. These are classical foundations (**??**), not a claim of a new fixed-point principle. Pointwise transition continuity with an unbounded innovation support is insufficient for arbitrary monetary risk; the exact essential-uniform alternative is stated in the proof.

The original investment primitives satisfy these conditions directly: $F(x,a,z) \in [1/32, 27/32]^d$, $1/8 \le A(x) \le 1/4$, and all costs are bounded. Their joint curvature gives V_1 state strong-convexity modulus $(4 - 5\beta/4)/d$ and root action modulus two. Both dates therefore have unique continuous constrained optimal actors. Proposition S.10 also derives the continuous value gradient with its capacity multiplier, the original-risk root derivative, and the actual observation/feasibility allowance. Differentiating a rounded actor is not used to establish any of these claims. The supplement records the full smooth critic–actor graph and distinguishes its gradient from a frozen-target update.

The separate Brownian recursive-utility bridge has its own existence and verification assumptions. Proposition S.11 supplies the stopped-domain version, charging the original stopping boundary and terminal condition. Its Lipschitz driver can depend on the backward value, in which case cash invariance is not automatic. The continuous-time stability factor and smooth critic graph are therefore verified within that original recursive criterion.

6. LEARNING THE MARGINAL CONTINUATION OF THE ORIGINAL BELLMAN OPERATOR

The preceding comparison methods certify an acquired actor. We now change what the critic learns. The target remains the original two-date investment problem in Section 5, with its continuous innovation law, capacity, discount and nested AVaR preference. The construction first evaluates the original risk operator and then learns its action marginal. In particular, it does not replace the optimizing risk density by the density selected by a fitted terminal continuation.

Write $a = A(x)v$, $v \in [0,1]$, and define

$$\begin{aligned} R_\alpha(x,v) &= \beta \rho_\alpha\{V_1(F(x,A(x)v,Z))\}, \\ Q_x(A(x)v) &= S(x) + h(A(x)v) + R_\alpha(x,v), \\ H_\alpha(x,v) &= A(x)^{-1}\partial_v R_\alpha(x,v). \end{aligned} \tag{20}$$

Thus H_α is a marginal continuation cost per unit of the original action. The known stage cost is retained analytically. The original optimal last-date value $V_1 = T_1 g$ is the target of the paid acquisition calculation. Deployment still uses the same

measurable last-date controller as the retained study. Its uniform guarantee gives

$$J_x(a) = Q_x(a) + \eta_x(a), \qquad 0 \le \eta_x(a) \le \delta := \beta 2^{-20}. \tag{21}$$

No differentiability or convexity of η_x is assumed.

Derivative-based economic dynamic programming, derivative supervision and decision-aware learning have established precedents (**????**). The action-null quotient principle is also retained from R52, and the constrained directional regret bound from Theorem S.1. R67 already supplies a certified fixed-feature readout and a dual gap. The additional construction here supplies regularity constants for the original recursive AVaR target, outward derivative labels made from paid finite-risk values, a finite-class training certificate, and a feature-aware extension from those labels to off-design decisions. These ingredients apply to conventional and neural representations with the same information and feature restrictions.

6.1. *Regularity of the original integrated risk marginal*

Throughout this section $d = 2$, $R = 1/32$, $\beta = 15/16$, $\alpha \in \{1/2, 3/4\}$, and $t = 1 - \alpha$. Retain the original state-gradient constant from Theorem S.4:

$$\begin{aligned} M_\alpha &= \frac{1413}{128} + \beta\left\{\frac{127}{16} + \frac{19625\alpha}{20480t}\right\}, \\ m_z &= \frac{181}{64}, \qquad C_\alpha = \frac{4}{5}M_\alpha, \\ D_\alpha &= \frac{5M_\alpha}{16} + \frac{C_\alpha^2\alpha}{m_z t}, \qquad L_\alpha = 5 + \beta D_\alpha. \end{aligned} \tag{22}$$

All these constants are derived from the original primitives. They are not fitted from the acquisition or evaluation sample.

THEOREM 6—An original-risk marginal with explicit boundary control: *The function H_α in (20) is continuous on $\Omega = [0,1]^2 \times [0,1]$, with the inward derivative convention at the action endpoints. For each state, Q_x is continuously differentiable and 2-strongly convex on $[0, A(x)]$, and*

$$|Q_x'(a) - Q_x'(b)| \le L_\alpha |a - b|.$$

Consequently its normalized action curvature lies between $2A(x)^2$ *and* $L_\alpha A(x)^2$. *For* $u = (x, v)$ *and* $u' = (x', v')$,

$$|H_\alpha(u) - H_\alpha(u')| \le \Lambda_\alpha \|u - u'\|_2, \tag{23}$$
$$\Lambda_\alpha = \beta \left\{ \frac{441 M_\alpha}{1024} + \frac{147 M_\alpha^2 \alpha}{160 m_z t} \right\}.$$

These statements include a minimizing central risk interval at either endpoint of its feasible set and a binding economic capacity.

The proof uses the original M_α-Lipschitz gradient of V_1 and the m_z-strong convexity of $z \mapsto V_1(F(x, a, z))$. A central interval of length $2R\alpha$ has a unique minimizing location. Its constrained first-order inequalities, rather than an assumed interior root, control changes in the adverse tail. That step is essential: controlling a fitted terminal continuation in one action direction alone need not control which risks receive AVaR weight.

6.2. *Paid derivative labels and finite-class training*

At a declared acquisition site (x, v), set $a = A(x)v$. Suppose an original-Q_x oracle returns outward value intervals at a and the available inward probes $a \pm d$, where $d > 0$ is fixed before their outcomes and each queried action is feasible. Write $\mathcal{I}_0, \mathcal{I}_+, \mathcal{I}_-$ for these intervals, and use outward interval subtraction and division to form

$$\mathcal{S}_+ = (\mathcal{I}_+ - \mathcal{I}_0)/d, \qquad \mathcal{S}_- = (\mathcal{I}_0 - \mathcal{I}_-)/d, \qquad \mathcal{S}_c = (\mathcal{I}_+ - \mathcal{I}_-)/(2d).$$

PROPOSITION 7—Finite-risk secants enclose the original marginal: *Every available interval in the following list contains* $Q_x'(a)$:

$$\begin{aligned} \mathcal{G}_+ &= \mathcal{S}_+ + [-L_\alpha d/2, -d], \\ \mathcal{G}_- &= \mathcal{S}_- + [d, L_\alpha d/2], \\ \mathcal{G}_c &= \mathcal{S}_c + [-(L_\alpha - 2)d/4, (L_\alpha - 2)d/4]. \end{aligned} \tag{24}$$

The central interval requires both probes. The forward interval alone is valid at $a = 0$, and the backward interval alone at $a = A(x)$. Their outward intersection, followed by subtraction of the known $h'(a)$, gives a valid label $[\ell_i, u_i]$ for $H_\alpha(x, v)$.

The experiment uses the prospectively specified normalized probe 2^{-12}, hence $d = A(x)2^{-12}$. Its five main action sites are $v \in \{0, 1/4, 1/2, 3/4, 1\}$. The three interior sites buy three original-Q values each, and the endpoints buy two each: there are thirteen paid contexts at each state. The nested state meshes have 5^2 and 9^2 sites, giving 325 and 1053 contexts per risk level and standalone acquisition budget. The protocol specifies how each continuous-risk interval is constructed and charges all optimal-continuation solves, probes and enclosure work. The allowance δ in (21) does not enter an optimal-Q derivative label; it enters the subsequent implemented-policy statement once.

For a declared family of continuous piecewise-polynomial features, use the anchored critic

$$\widehat{R}_\theta(x, v) = \widehat{R}_0(x) + \sum_{j=1}^{p} \theta_j \{\phi_j(x, v) - \phi_j(x, 0)\}. \tag{25}$$

The implementation stores an approximation $\widehat{Q}_0(x)$ to $Q_x(0)$. In (25) this means $\widehat{R}_0(x) = \widehat{Q}_0(x) - S(x)$, so the full action critic is $\widehat{Q}_0(x) + h(A(x)v)$ plus the learned anchored difference. The state-only anchor affects a displayed value level and cancels from every current action comparison. At a label site, let $A_{ij} = A(x_i)^{-1} D_v \phi_j(x_i, v_i)$. Include both feasible one-sided rows at an interior kink and only the inward row at $v = 0$ or $v = 1$. The same label is attached to each row. Let the coefficient box $\Theta = \prod_j [\underline{\theta}_j, \overline{\theta}_j]$ be fixed before acquisition. The action-marginal training objective is

$$T(\theta) = \max_i \{A_i\theta - \ell_i,\ u_i - A_i\theta\}, \qquad T_* = \min_{\theta \in \Theta} T(\theta). \tag{26}$$

It is the largest possible node error over the retained outward labels, in original-action units. The value-loss control uses the same box, features and complete paid bundle, and fits the anchored values by bounded least squares. A smaller value loss is not presumed to minimize (26).

PROPOSITION 8—A checkable global training gap in the declared box: *The minimum T_* exists. Any exactly checked $\widehat{\theta} \in \Theta$ gives the primal upper bound $P = T(\widehat{\theta})$. Let $y_i^+, y_i^- \geq 0$ be rational weights whose combined sum is one, and set*

$$r = \sum_i (y_i^+ - y_i^-) A_i, \qquad B_y = -\sum_i \ell_i y_i^+ + \sum_i u_i y_i^- + \sum_j \min\{\underline{\theta}_j r_j, \overline{\theta}_j r_j\}.$$

Then

$$\max\{0, B_y\} \leq T_* \leq P, \qquad 0 \leq P - T_* \leq P - \max\{0, B_y\}. \tag{27}$$

In particular, no exact dual stationarity equations or solver status are needed for the lower bound. Exact verification must bind the actual stored coefficients, feature rows, labels and box.

The declared box is $[-128, 128]$ except for the nonnegative quadratic curvature coefficient and the nonnegative nonlinear ReLU or matched-hinge coefficients. A signed coordinate contributes $-128|r_j|$ to B_y; a nonnegative one contributes $\min\{0, 128 r_j\}$. The quadratic, polynomial, spline, tensor-grid and matched-hinge controls face the same acquisition interface. A coordinate-loop hinge carrying the same features and coefficients represents the same function as its ReLU encoding. This experiment can identify the consequence of a learning loss or encoding cost, not a neural-specific approximation advantage from that equality.

6.3. *From finite labels to an off-design policy certificate*

A node training certificate does not supply uniform derivative accuracy. To make the remaining calculation explicit, take a finite cover $\mathcal{C}$ of Ω. For each cell C, construct intervals

$$d_j(C) \supseteq \{A(x)^{-1} D_v \phi_j(x, v) : (x, v) \in C\},$$

including every relevant one-sided value. Let $[\underline{h}_\theta(C), \overline{h}_\theta(C)]$ enclose $\sum_j \theta_j d_j(C)$ by interval arithmetic, and let $r_i(C) \geq \sup_{u \in C} \|u - \xi_i\|_2$, where $\xi_i = (x_i, v_i)$ is the parameter node. Define

$$\underline{H}(C) = \max_i \{\ell_i - \Lambda_\alpha r_i(C)\},$$

$$\overline{H}(C) = \min_i \{u_i + \Lambda_\alpha r_i(C)\},$$

$$\epsilon(C) = \max\{0, \overline{h}_\theta(C) - \underline{H}(C), \overline{H}(C) - \underline{h}_\theta(C)\}. \tag{28}$$

Using only a declared subset of label sites in the maximum and minimum remains valid. Empty target intersections indicate a failed numerical enclosure and do not authorize selection of a favorable endpoint.

THEOREM 9—Feature-aware continuation and decision control: *Suppose the label and feature-range inequalities above are valid. For each cell C, the original-action marginal error of (25) is at most $\epsilon(C)$, including one-sided limits. Fix a state x, and let $\epsilon_x(v)$ be any measurable choice of a valid cell bound at (x, v). For feasible original actions a, b, put*

$$\mathcal{D}_x(a,b) = A(x) \int_{\min(a,b)/A(x)}^{\max(a,b)/A(x)} \epsilon_x(v)\, dv,$$

$$\widehat{q}_x(a) = S(x) + h(a) + \widehat{R}_\theta(x, a/A(x)). \tag{29}$$

Then

$$|Q_x(b) - Q_x(a) - \{\widehat{q}_x(b) - \widehat{q}_x(a)\}| \le \mathcal{D}_x(a,b), \tag{30}$$

$$J_x(b) - J_x(a) \le \widehat{q}_x(b) - \widehat{q}_x(a) + \mathcal{D}_x(a,b) + \delta,$$

$$0 \le J_x(a) - V_0(x) \le \delta + \sup_{0 \le b \le A(x)} \{\widehat{q}_x(a) - \widehat{q}_x(b) + \mathcal{D}_x(a,b)\}. \tag{31}$$

Any certified upper bound on the final supremum remains valid. These conclusions do not require a convex fitted critic.

The finite cover pays for the actual feature geometry. An anchored ReLU feature has derivative zero or its declared action coefficient; both sides enter a crossing cell. Polynomial and spline pieces have computable polynomial derivative ranges. Tensor hats have one-sided action derivatives at their knots. Dividing their numerator bounds by the positive capacity interval is valid because $A(x) \ge 1/8$. The supplement gives constructive range rules for every declared family. Consequently (28) does not infer a smooth fitted derivative from smoothness of the original target.

The bound can be conservative at the fixed training meshes. Unresolved cells, paid refinements and fallback original-risk qualification remain part of the numerical record. No off-design certificate is credited merely because a finite-dimensional fit has a small node loss.

PROPOSITION 10—Capacity-sensitive actor extraction: *Suppose a feasible returned action a has an outward original derivative interval $[g_-, g_+]$. Define*

$$\mathcal{K}_x(a;g) = -\min_{d\in[-a,A(x)-a]}\{gd + d^2\},$$
$$K_x(a) = \max\{\mathcal{K}_x(a;g_-), \mathcal{K}_x(a;g_+)\}. \tag{32}$$

Then

$$Q_x(a) - V_0(x) \le K_x(a), \qquad J_x(a) - V_0(x) \le K_x(a) + \delta, \qquad |a - a_0^*(x)|^2 \le K_x(a).$$

The scalar minimum is attained at $\mathrm{proj}_{[-a,A(x)-a]}(-g/2)$. *At* $a = 0$, $g_- \ge 0$ *implies* $K_x(a) = 0$; *at* $a = A(x)$, $g_+ \le 0$ *implies* $K_x(a) = 0$. *If the fitted actor has a checked constrained stationarity residual at most* r, *and every required one-sided marginal error is at most* ϵ, *the interval supplied by that error bound can be chosen so that*

$$K_x(a) \le (r + \epsilon)^2/4.$$

This is the retained directional strong-convexity bound applied to a newly acquired and extended marginal label. It makes a capacity action with the correct marginal sign economical to certify. A numerical polynomial root or an optimizer status alone does not supply the stated residual; the returned action, piecewise derivatives and rounding must be checked or covered by the original certificate service. Similarly, neither a training-gap certificate nor a certified action identifies a computational adoption region without the full acquisition and decision-service costs and the common-information comparisons.

6.4. *The anchor that a recursive Bellman operator must retain*

An action-independent level cancels at its current state. It need not cancel when that learned value is exported as the next continuation. The original investment

transition gives an exact example. If $W_1(y) = V_1(y) + \gamma(y_1 + y_2)$, the zero-sum shock and cash invariance imply

$$Q_x^{W_1}(a) - Q_x(a) = \beta\gamma\{F_1(x,a,0) + F_2(x,a,0)\}.$$

The added root marginal is $3\beta\gamma/4$. The same state-dependent level could leave a last-date action minimization unchanged. Thus discarding that anchor would change the recursive decision problem.

The complete-value statement in Theorem S.22 supplies the required account. Let W_{t+1} be an exported continuation with signed uniform errors $E_{t+1}^- \le W_{t+1} - V_{t+1} \le E_{t+1}^+$. Suppose the acquired action critic is compared with the original Bellman objective using that W_{t+1}, and its retained anchor and action-contrast errors give full-level error bounds $d_t^- \le \widehat{Q}_t - Q_t^{W_{t+1}} \le d_t^+$. If the exported value has an additional checked minimization error at most ζ_t in absolute value, then a monetary recursive risk map gives

$$\begin{aligned} E_t^- &= d_t^- + \beta_t E_{t+1}^- - \zeta_t, \\ E_t^+ &= d_t^+ + \beta_t E_{t+1}^+ + \zeta_t. \end{aligned} \tag{33}$$

A local certified action gap g_t under $Q_t^{W_{t+1}}$ and future policy bound P_{t+1} give

$$P_t \le g_t + \beta_t(E_{t+1}^+ - E_{t+1}^-) + \beta_t P_{t+1}.$$

In the present paid two-date construction the root target uses the original V_1, so its continuation approximation term is zero; the unchanged terminal controller contributes $\beta\varepsilon_1 = \delta$. The supplementary theorem separately states the bound for a noncash-invariant recursive preference under an explicit continuation Lipschitz assumption. Such a preference does not inherit the monetary shift identity by notation.

7. AN INDEPENDENT COMPARISON OF LEARNING LOSSES

This section is the prospective manuscript component at source freeze F. The R121 economic acquisition and independent evaluation have not yet been executed at that boundary. The completed author source replaces this paragraph with the

actual execution identities, complete numerical tables and economic interpretation. No earlier experiment supplies R121 outcomes.

The original model is (13)–(14) at $d = 2$, two dates and risk levels $1/2$ and $3/4$. Two endpoint state grids have 5^2 and 9^2 states. Each buys five normalized-action derivative contexts and their fixed feasible probes, or thirteen original-value contexts per state. Each budget is actually acquired in a separate process. Every method within it receives the same complete information bundle and is charged for its standalone acquisition.

The seven stored families are a convex quadratic, an anchored ReLU critic, its exactly corresponding conventional hinge encoding, a tensor cubic spline, unrestricted polynomials of degrees two and three, and a multilinear grid. The VALUE and MARGINAL fits use the same coefficient box and features within a family. VALUE uses every paid value, including the probes. MARGINAL uses the outward original-action labels. The online baseline computes its proposal directly from the original model. Every proposal faces the same complete original-law certificate, refusal, conventional fallback, integration refinement and rational totalization.

The source/protocol commit precedes all acquisition. The acquired model commit precedes a single new operating-system entropy draw. The state commit precedes every outcome. There are 58 policy cells, 3,712 primary singleton decisions and 9,280 decisions in five serial fresh timing processes. The independent economic reconstruction evaluates all distinct final-policy and raw-proposal keys with the same rational design, and the separate training audit reconstructs the actual algebraic feature rows and finite-box gaps. The 64 states, rather than risk cells or repeated clocks, are the independent statistical units. The complete prospective design and reporting family are retained in `PROTOCOL121.md`, `INDEPENDENT_PROTOCOL121.md` and `REPORTING_SPEC121.md`.

8. ECONOMIC USE AND THE REMAINING RESEARCH TASK

The role of a Neural Bellman Operator is to acquire and reuse continuation information that changes feasible economic decisions at an acceptable total cost. The marginal construction gives that objective a precise training and verification account. It also exposes two costs that cannot be removed by renaming a fitting

statistic: integration must track the original risk operator, and an exported value must retain its state-dependent level.

The acquisition budgets in this experiment change information density and representation size within the same two-state economy. They do not establish scalability in horizon, state dimension or innovation dimension. The retained theorems describe those separate objects and their assumptions; their historical experiments and unresolved regions remain in the development companion. The stopping, accuracy-allocation and risk-information questions therefore remain part of the NBO programme under the original economic criteria.

A neural-specific advantage requires a distinction that a conventional method with the same information, features, feasible policies and resources cannot reproduce. The exactly matched hinge control makes one such boundary explicit. Conditional adoption additionally requires the original objective gain to pay for all acquisition and complete-service work at a stated price and volume. These are empirical and mathematical requirements for an economically useful NBO, not consequences of a successful optimization status or a smaller internal loss.

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